

THROTTLE CONTROL OF HIGH-POWERED LIQUID ROCKETS USING ELECTRONIC-REGULATORS

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GENERATIVE AI STATEMENT: The AI ChatGPT was occasionally used by the student in initial stages to explore concepts and ideas, but no AI prose is used in this document. AI was used to generate some BibTex and LaTeX formatting. AI was used to generate some code for graph plotting and formatting, but none of the controller or modelling code. The student reviewed and edited the content as needed and takes full responsibility for the content submitted for assessment.

ABSTRACT

This paper presents the design, implementation, and planned testing of a new controller and system architecture for electronic pressure regulation in a student liquid-bipropellant rocketry team. A predictive strategy is determined to enable robustness to engine restarts, and a new method of gain scheduling is used. Future experimental testing is laid out, and a Simulink model is developed for evaluating the model. Results demonstrate the feasibility of the proposed approach in allowing new capabilities, but further empirical testing is required before the system can fly.

1 INTRODUCTION

1.1 Background

In pressure-fed rocket engines, a high pressure inert gas is used to create a pressure gradient into the combustion chamber through the tanks, providing a flow of fuel and oxidiser into the combustion chamber.

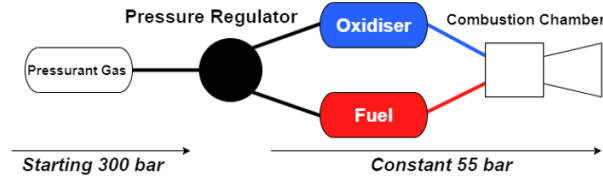


Figure 1: A diagram showing the placement of a pressure regulator within a typical feed system.

In order to keep this flow steady, it must be controlled.

Often in lower-cost applications, a mechanical pressure regulator is used. These reduce the pressure through the outlet via a spring-diaphragm mechanism, as pictured in Figure 2. By adjusting the compression on the spring by turning a dial, the preset pressure can be adjusted.

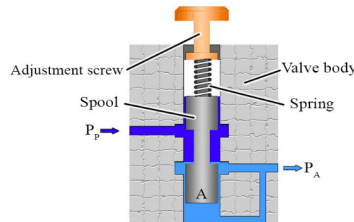


Figure 2: A mechanical pressure reducing valve. Image source: [1]

The reliability of mechanical pressure regulators is useful, hence they find use in propulsion systems for cheaper telecommunications satellites and student rocketry teams. However, they also come with numerous drawbacks.

The inability to vary the desired pressure during operation means it is impossible to throttle the engine. This is a requirement for maximum efficiency during ascent, or to allow propulsive landings. They also

suffer from a phenomenon known as 'droop', where an increase in flow rate causes a steady state error in outlet pressure. Finally, they are simply heavy. By nature of their mechanism, they require heavy springs and dense housings.

The alternative to this is an electronic pressure regulator, or '*E-Reg*'. In these systems, a closed-loop control system measures the upstream and downstream pressure, and finely actuates a valve between to control this pressure drop. As a fundamentally non-linear, time-variant system, the typical PID is not most suitable for this problem. Previous approaches involve time-based gain scheduling with model predictive control [2] and ullage-volume estimations for gain scheduling with a PI controller [3]. Lee and Roy also demonstrate the capability for throttling provided by a quad E-Reg system in paper [2].

1.2 Research Objectives

In this paper a new architecture of electronic pressure regulator control is proposed, designed for the ball-valve-servo E-Reg hardware produced by Hypower Bristol. The aim is to produce a robust control method capable of variable throttling, and stable performance throughout an engine stop and relight procedure.

The proposed regulator architecture is a dual E-Reg system with both valves upstream, which has the same capability for pressure regulation and throttle control as Lee and Roy in Paper [2] without needing as much hardware. A new controller architecture is also proposed, with an adaptive gain based on upstream tank pressure, and simple logic (feedforward) for the relight procedure. The solution is analysed with

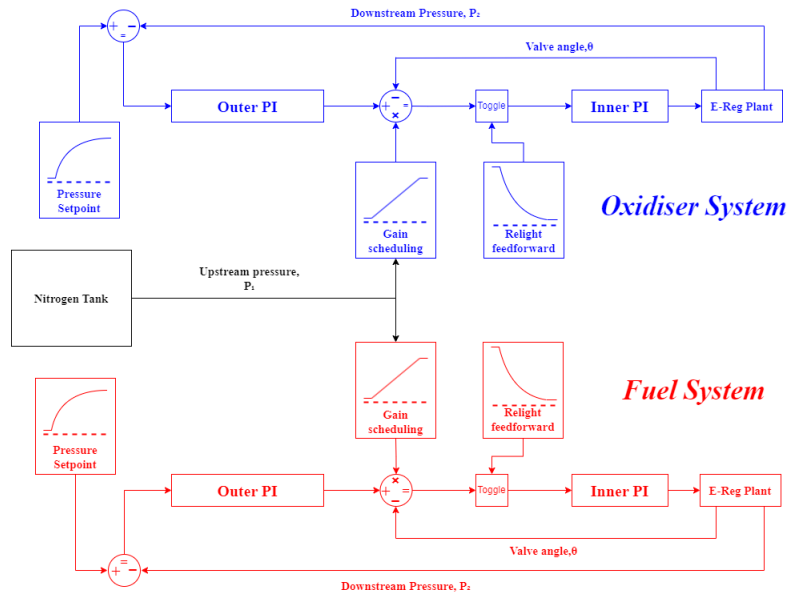


Figure 3: The dual-cascade controller architecture showing the 2 controllers with shared non-linearity adjustment based on upstream pressure, and interrupting predictive logic for the relight procedure.

respect to integration into Hypower Bristol's launch vehicle. Figure 3 depicts this architecture. The advantages of this adaptive dual architecture when compared to previous approaches are:

- Capability for independent throttling. This also allows for the OF ratio to be changed even during a burn.
- Robustness to relight. This year, Hypower Bristol are pursuing a relight procedure during flight. This is incompatible with previous time-based gain scheduling methods.
- Unlike in England's approach in paper [3], gain scheduling is not based on a tank plant model but on a single measurement, potentially making it more reliable.

In Table 5, requirements for the controller performance are set, determined to improve upon the performance of Hypower's previously tested mechanical regulator. The requirements are to be met for a typical Hypower launch case.

A project plan for the full development of the E-Reg system across 2025 was determined as follows:

Requirement	Value
Inlet pressure range	120-300 bar
Outlet pressure range	30-55 bar
Max. transient error	3 bar
Max. Steady-state error	1 bar
Throttle capability	80%
Robust to relighting procedure.	

Table 1: Requirements for the system controller.

1. Perform first air/water experiment to characterise valve flow behaviour. (*Jan-Feb*)
2. Integrate empirical data into Simulink model of feed system, for initial controller tuning. (*Feb*)
3. Repeat experiment with active control to verify controller performance, adapt parameters as required. (*Mar*)
4. Complete RP3 Hand-in. (*Apr*)
5. Complete autocoding, hardware development and integration into Hypower feed system. (*Jun*)
6. Perform cold-flow and hotfire tests in collaboration with industry partners and Hypower. Demonstrate reliability and throttle capability¹. (*Jun-Jul*)
7. Successfully fly system at EuRoC 2025. (*Oct*)

1.3 Hypower Bristol system

This project is aimed to integrate into Hypower Bristol’s 2025 EuRoC² launch vehicle, named *Warwing*, which is currently under development. This is a pressure-fed system with a thrust of 5000N, and a target apogee of 3000m. The fuel and oxidiser respectively are ethanol and nitrous oxide. The E-Regulator itself consists of a simple COTS ball valve connected via two gears to a servo, secured by an aluminium bracket. This hardware is identical to other actuated valves in the launch vehicle (used to control filling and venting of propellants). Angle sensing is provided by a Hall effect sensor mounted on the valve stem.

1.4 Experimental Overview

In order to develop this controller and verify its performance to a flight-safe level, empirical data is required. First, to properly characterise the flow rate through the valve for different pressure deltas and opening angles, and secondly to verify the performance of the controller under real conditions, and adjust the controller parameters as needed. For this, a pressure experiment was developed.

Unfortunately, the author was unable to perform the experiment within the required timeframe, due to the hazardous pressures involved. Hence, this remains future work, and the methods are nonetheless described.

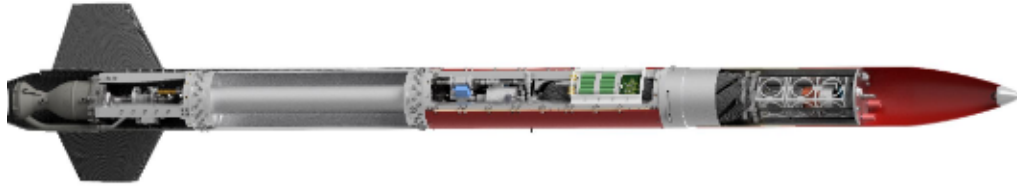
1.5 Theoretical Overview

The aim is to develop a controller that can effectively control the pressure in the lower tank, including during relight and throttling. A MATLAB/Simulink model is used to simulate the feed systems of the Hypower rocket during a typical launch. The proposed controller and E-Reg architecture is integrated into this.

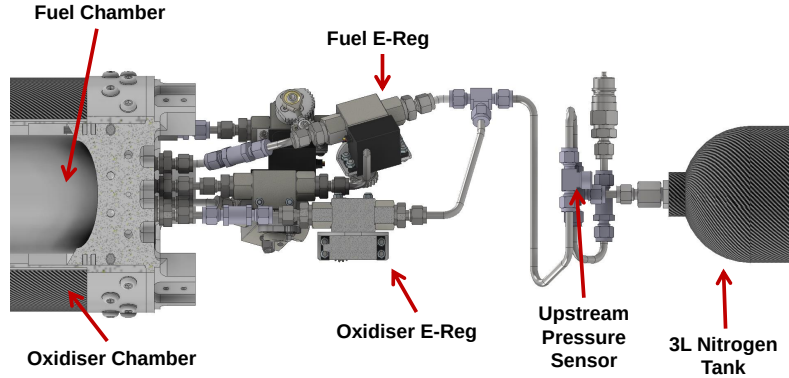
This allowed the controllers to be tuned via a heuristic approach, in order to develop some initial controller parameters and verify the viability of the architecture in the Hypower system.

¹A design gate at this point will determine whether E-Regs will be used for flight or delayed for 2026. The system is designed to function both with E-Regs fitted or the previous mechanical regulator.

²The European Rocketry Challenge, an undergraduate competition hosted by the Portugese Space Agency.



(a)



(b)

Figure 4: (a) A render of the *Waxwing* launch vehicle. (b) The annotated upper feed system, containing the E-Regs.

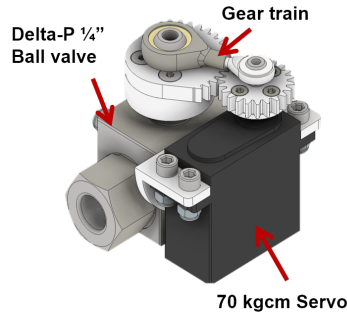


Figure 5: The E-Regulator hardware.

2 THEORETICAL METHODOLOGY

2.1 Controller details

Feedforward

In order to allow the outer controller to effectively manage the disturbances caused by the relight procedure, feedforward control is used. Throughout the 1-second long relight procedure, the run valve (allowing fuel to travel to the engine) is closed. This means the pressure in the fuel tank cannot decrease. If the controller dealt solely in feedback with this, there would be a potentially large overpressure in the tank due to the delay, and the integral term would saturate, leading to an overcorrection once the relight finishes. Since the time at which the relight starts and finishes is known, the controller can switch to an open-loop mode and pre-empt the disturbance by closing the E-Reg (which takes a known duration based on current position) in advance.

Adaptive Gain Scheduling

The response of the fuel tank pressure to E-Reg actuation is highly dependent on both the ullage fraction

in the downstream tank and the pressure delta to the nitrogen tank. This is shown in Figure 6. In effect,

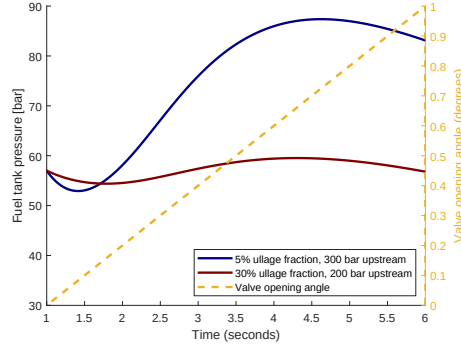
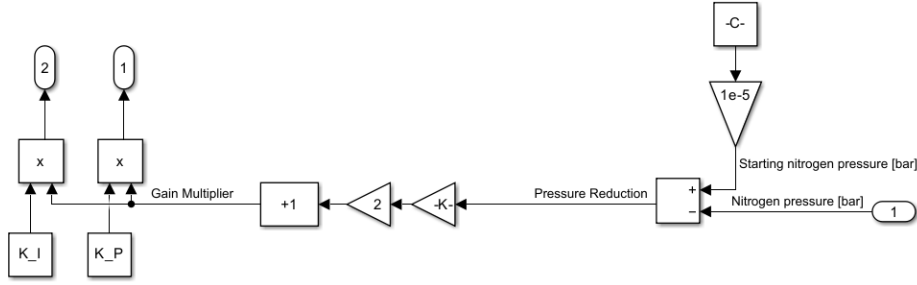


Figure 6: The response to the same E-Reg programme for two different cases.

the system behaves more like the blue response at the start of operation, and more like the red response just before engine cutoff. To account for this adaptive gain scheduling is implemented, which scales the gains of the outer controller based on a function of the pressure in the nitrogen tank.



$$Mult = 1 + 2.5(-\Delta P_{N_2})$$

Figure 7: The Simulink subsystem that implements gain scheduling, along with its equation.

Various functions were tried, including quadratics and exponentials, but the simple linear equation above was found to effectively mitigate the changing sensitivity without making the controller difficult to tune for all scenarios. It was also found that a constant derivative gain is preferred, as scaling this caused the controller to consistently overcorrect.

Inner Controller

This is a much simpler controller, consisting of a PI (derivative omitted to mitigate noisy data of the Hall effect sensor) and no feedforward or adaptive gain. It is used solely to order the servo to the position demanded of it by itself and the outer controller.

This controller was tuned first (using heuristic methods) to provide a baseline for the rest of the system.

2.2 Modelling

2.2.1 Flow Solver

The fluidic modelling and flow solver used is produced by ICL Rocketry[4], which numerically solves a system of equations based on ISO-6358-3[5] (compressible flow through a component) along with Bernoulli's equation applied at orifices, with the constraint of conservation of mass, creating an effective model for an adiabatic compressible gas/liquid interface and tank plant. Since it outperformed (both in accuracy and computational performance) the author's own attempt, it was used in this project with permission.

2.2.2 Valve-Servo subsystem

Backlash Hysteresis

The two gears that provide the mechanical connection between the valve and servo do not provide an ideal connection. *Backlash* is one effect that introduces non-linearity, causing a discontinuity between the

servo angle and actual valve opening position. This effect that manifests as a slight slack that occurs when the servo changes direction. This has been mitigated with the addition of a rose joint to hold the gears in tension, and the use of a hall-effect sensor on the valve stem itself allows an accurate measurement of the valve angle. Backlash is however intrinsic to geared systems and cannot be eliminated entirely. The angular backlash was experimentally determined to be $3^\circ \pm 0.5^\circ$, and the Simulink model was modified to detect and account for the hysteresis response when the servo changes direction, as shown in Figure 8. This removes the effect of small-magnitude, high-frequency corrections, hence the controller is tuned

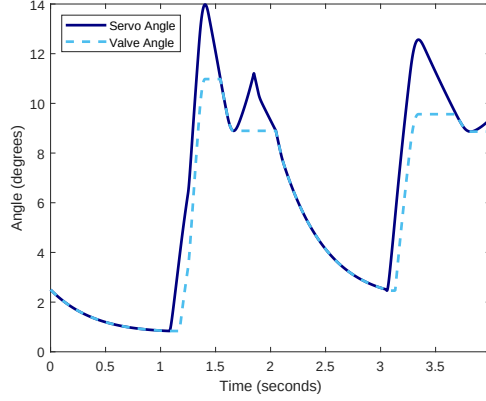


Figure 8: 4 seconds of arbitrary servo and valve movements. Note the "flattening" hysteresis response as the slack in the system is picked up. The gear ratio has been set to 1 for ease of comparison.

with avoiding these in mind. Furthermore, a low-pass filter is used on the valve-angle signal. This will likely become more important on the physical controller, as the hall-effect sensor is prone to fluctuation.

Valve Flow Coefficient

Yet another non-linear effect in this system is the effect of valve opening angle on the actual flow rate and pressure drop this causes. This is typically described in terms of a function of *Flow Coefficient*, K_V versus opening angle. The Flow Coefficient is described as follows.

$$K_V = Q \sqrt{\frac{SG}{\Delta P}} \quad (1)$$

Where SG refers to specific gravity of the fluid, Q is the flow rate and ΔP the pressure drop.

Since this data is not published for the valve model used (except for the fully open case where $K_V = 1^3$, it must be experimentally determined. However, the experiment remains yet to occur so an approximation was used instead, based on a geometric model. This will be replaced at a later date.

A MATLAB script was written to calculate the varying intersection area of two circles of equal radius

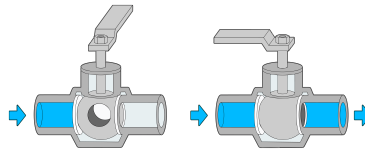


Figure 9: The mechanism of a ball valve. Image credit: [6]

moving towards each other⁴, to give an estimate of the flow area. An empirical correction factor was applied to scale the result to match the known K_V when in the fully open position, and account for the effects of the varying flow path angle. The results of this are shown in Figure 10, along with empirical data from a different ball valve. From inspection, the curve's shape correspond well, turning from convex to concave at nearly the same point of rotation. Hence this model can be considered adequate until empirical testing can better establish the true relationship.

³Under standard water flow conditions

⁴Geometrically, this shape is named a lens.

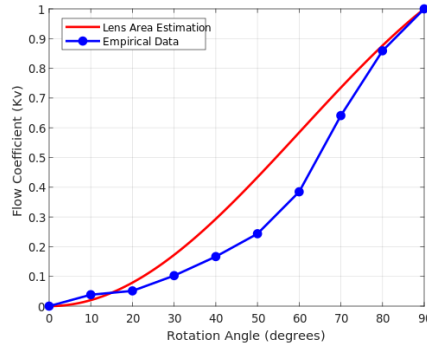


Figure 10: The normalised K_V versus valve angle of the theoretical model and empirical data for a different ball valve.[7]

2.2.3 Modelling Limitations

Two-Phase Oxidiser Modelling

Although the model as described can simulate a pressure-fed ethanol feed system, The oxidiser nitrous oxide is not a simple fluid to model. At the relevant temperatures and pressures, it is more accurately described as a system of fluids, as it exists in equilibrium with both liquid and vapour phases. As such, it is strongly non-ideal.⁵

Even modern models of bi-phase dynamics, such as modified Peng-Robinson equations of state, have large discrepancies from real flows particularly closer to the critical point[8] (as this system is). This is before the addition of nitrogen as a pressurant mixing with this complicates things even further. Models for the self-pressurisation case are viable, but the elimination of the effect of pressurant mixing makes such models useless for this project.

As such, it was decided that the development of an accurate two-phase oxidiser mixing model was outside the scope of this project. Although the controller architecture is designed for robustness to diverse non-linearity, empirical verification of the oxidiser controller performance is required before the system can be cleared for flight.

Time Delay

The time delays from circuitry and sensing are taken from the relevant documentation of the sensor where possible, but are occasionally estimated.

Bandwidth

The speed of valve actuation was measured to be $180^\circ/s$, but this was only possible to measure under no pressure load. It is possible that in real operation up to 300 bar this value will change, effecting the system response and particularly the feedforward logic.

⁵One effect of nitrous oxide's strange behaviour is its tendency to *self-pressurise*, a very useful property as it can supplant the need for a pressurant gas. Unfortunately, in this case the flow proves too unsteady for stable combustion, but has application for hybrid motors.

3 EXPERIMENTAL METHODOLOGY

3.1 Pressure Safety and regulation

The high pressures present in any feed system or related experiment are inherently hazardous. For a (rather violent) point of comparison, the energy stored in the high pressure nitrogen tank is approximately comparable to the equivalent kinetic energy of 3 .50 calibre bullets in flight.

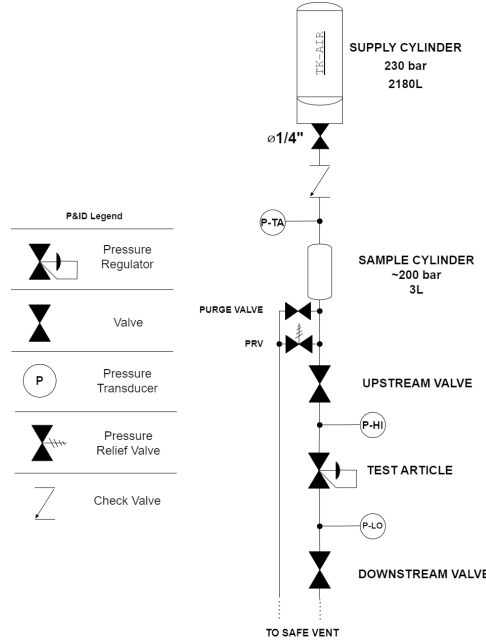


Figure 11: Plumbing & Instrumentation Diagram for the experiment

As such, it is important to ensure any pressure experiment is carried out safely. In the UK pressure systems are governed by the Pressure Systems Safety Regulations 2000.(PSSR)[9]⁶ There are 4 main types of injury from pressure systems shown in Table 2 which were kept in mind when designing the system.

Hazard	Injury(s)	Cause(s)	Mitigation(s)
Projectiles	Pressure injection, kinetic	Fast pressurised failure, loose fittings	Securing apparatus
Jets	Pressure injection	Loose fittings, insecure apparatus, trapped pressure	Proof testing, protective equipment
Whips	Pressure injection, kinetic whipping	Unfastened hose	Whip checks
Blast	Overpressure, kinetic, secondary failures	Brittle failure at pressure	Proof testing, secure environment

Table 2: The main risks from pressurised systems.

Securing apparatus was designed to make this experiment safe using properly rated equipment, and test plans and procedures were written. This allowed the experiment to be tolerant of test item failure and still be made safe. Any opportunity for trapped pressure was mitigated with the addition of redundant Swagelok Pressure Relief Valve and a manual purge valve, and a check valve prevented backflow into the filling system. This is shown in Figure 11.

⁶Although Exception 6 releases research experiment apparatus from these rules, it is stated that "Anyone relying on this exception should be able to justify their reasons for non-compliance and any failure to take the basic precautions required under the Regulations to prevent risk of injury from system failure." Hence all possible efforts are made to comply with the legislation.

3.2 Failure Modes and Effects Analysis

In order to characterise these risks, and develop methods of mitigating them, a Failure Modes and Effects Analysis (FMEA) was performed according to SAE J1739 [10] for each experiment iteration. The most recent such analysis is shown the Appendix B.

This demonstrated each of the most salient risks, which were addressed in turn until each risk was below the critical Risk Priority Number (RPN) of 150.

In addition, to understand the direct consequences of hazards occurring, a HAZOP (Hazard and Operability Analysis) was applied, shown in Appendix C.

3.3 Experiment Parameters

For the experiment described, various attributes were calculated as follows. Full details are listed in Appendix A. Most problematic of these is the loud venting noise, which is high enough to cause permanent

Attribute	Value
Maximum pressure of air in system	230 bar
Lowest pressure rating in system	334 bar
Equivalent air volume at ambient pressure (per test)	0.762 m ³
Total mass of air (per test)	0.784 kg
Equivalent air volume at ambient pressure (total supply)	552 m ³
Maximum flow rate	0.63 kg/s
Peak venting force	660N
Venting noise (at 3 metres)[11]	142 dB
Flow duration range	1.2 to 7s
Embodied energy (sample tank)	69 kJ

Table 3: Experiment Parameters

hearing damage. Although this can be mitigated for the operators with use of properly rated hearing protection, it requires a large safe cordon to prevent disruption and injury to passers-by. This was a large source of delay in finding a fitting site for the experiment.

4 RESULTS

4.1 Controller Performance

4.1.1 Case 1: Hypower launch programme

In order to test the theoretical performance of the controller, a mock Hypower launch programme was programmed as a variable setpoint.

- 0-1 seconds: Prelaunch, $P_{HP} = 300$ bar, $P_{Fuel} = 57$ bar
- 1-2 seconds: Initial launch, $P_{set} = 55$ bar
- 2-3 seconds: Relight procedure (throttle cutoff), $P_{set} = 55$ bar
- 3-4 seconds: Slow decrementing throttle, $P_{set} = 55$ bar $\rightarrow$ 53 bar
- 4-6 seconds: Fast incrementing throttle, $P_{set} = 53$ bar $\rightarrow$ 63.3 bar

The proposed controller outperforms a typical cascaded PID (with identical gains⁷) over all stages of flight. The simpler PID has slightly worse performance during the initial burn, but develops a large error due to its inability to predict the relight system behaviour. Finally, the lack of gain scheduling means the PID lags behind the setpoint significantly throughout the incrementing throttle manoeuvre.

The proposed controller does not suffer from these problems, and is within the requirements listed in Table 5. There remain some quirks. There is an underpressure of order 1 bar immediately prior to the relight sequence beginning. This is due to the feedforward necessitating E-Reg closure in spite of being at the setpoint before the engine cuts out.

⁷Heuristic tuning of the simple PID could not substantially improve performance over all stages of flight.

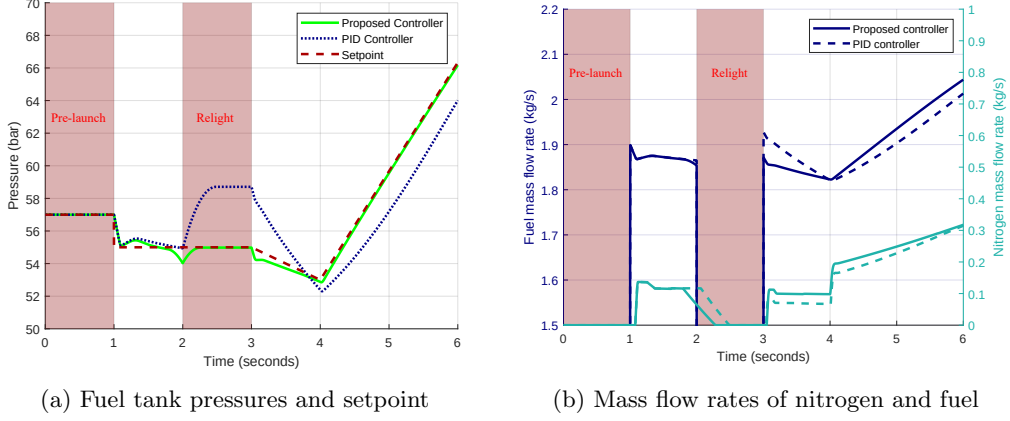


Figure 12: Time domain response of the proposed controller and a simple cascaded PID.

4.1.2 Case 2: Constant setpoint

In Figure 13, a constant setpoint pressure of 50 bar was also applied, to see how the system performed in a different case. Both controllers exhibit a small overshoot of 0.4 bar, but the proposed controller

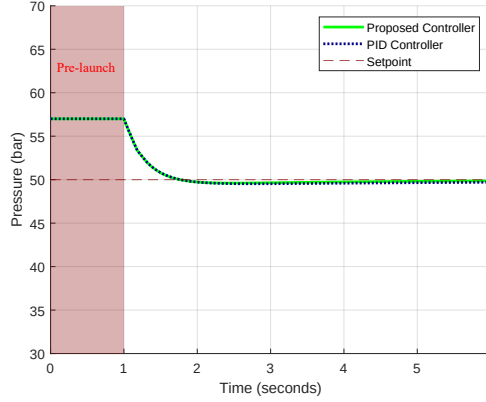


Figure 13: Time domain response of the proposed controller and a simple cascaded PID for the constant setpoint case

corrects this slightly quicker than the pure PID, due to gain scheduling increasing its sensitivity. The lack of any relight procedure means the feedforward does not come into effect in this case.

Here the proposed controller offers no significant advantages over a simpler cascaded PID, suggesting the architecture is only useful for launches where throttling and relights are present.

4.2 Literature Comparison

The proposed architecture and controller demonstrated some advantages and disadvantages over previous approaches in the literature in a qualitative sense⁸, particularly its ability to control throttling, and continue working throughout a relight procedure. It can demonstrate throttling capability like in Lee and Roy's quad-architecture model from paper [2], whilst using less hardware. However, it is unable to perform a rapid throttle decrement (due to the necessity of waiting for the fuel tank to depressurise), unlike Lee and Roy's[2] architecture which places more E-Regs downstream directly in the flow of fuel and oxidiser. Unlike England's architecture[3], the proposed system is capable of differential throttling. However, the more advanced logic present in the feedforward predictive control and ullage-space based gain scheduling provides a cleaner system response, at least in this testing without a nitrous-oxide model.

⁸Direct quantitative comparison is difficult due to the uniqueness of each system in the literature for which each controller is developed, and their differing roles and requirements.

Attribute	Value
Proportional gain K_{P1}	16
Integral gain K_{I1}	17
Derivative gain K_{D1}	8
Inner proportional gain K_{P2}	3.1
Inner integral gain K_{I2}	2.1
Adaptive gain function	$Mult = 1 + 2.5(-\Delta P_{N_2})$

Table 4: Controller Parameters

5 CONCLUSION

5.1 Objectives

Requirement	Value	Result
Inlet pressure range	120-300 bar	Met
Outlet pressure range	30-55 bar	Met
Max. transient error	3 bar	1 bar
Max. Steady-state error	1 bar	<0.5 bar
Throttle capability	80%	Met
Robust to relighting procedure.		Met

Table 5: Requirements for the system controller.

The proposed controller met the system requirements defined in Objectives within the MATLAB/Simulink environment. The model showed effective throttle control, relight robustness and adequate system response, but verification is required in an empirical manner.

The project plan points 1) and 3) defined in Section 1.2 were not achieved, due to the inability to perform the pressure experiment in time.

5.2 Future Work

5.2.1 Experimental

The clearest future work to be performed is the valve characterisation pressure experiment, which will allow a more accurate MATLAB/Simulink model of the system to be developed. Once new gains from this model are established, the E-Reg can be tested in active mode, followed by cold flows and hotfires in a full integrated test.

Currently, proof testing is planned to be performed at the fluid metrology company SM-Gauge, following which the other tests can take place, as safety documentation and test plans are finalised. If these are successful, and the model can demonstrate the same performance in reality, the E-Reg should be able to fly at EuRoC 2025 as part of Hypower Bristol’s entry.

Further, more speculative aims include combining the throttling capability of this controller with further control/actuation systems of TVC (Thrust-Vector Control), which could allow the launch vehicle to propulsively land.

5.2.2 Theoretical

The lack of empirical valve characterisation data limits the effectiveness of the Simulink model, at least with respect to determining accurate initial gains for the controller. This will be done soon. The model could also be improved with the addition of realistic sensor noise, and downstream pressure fluctuations from ignition and operation. Realistic pressure drops across the runs in the system would be useful in designing the optimum controller and integration with the rest of the system.

A greater limitation with the current model is the lack of an accurate oxidiser tank plant that can simulate a two-phase subcritical fluid interface with gas during draining flow. This is a substantial undertaking, but would be invaluable for making a full feed system model, and demonstrating capabilities like differential throttling.

Acknowledgements

I owe many people a debt for this project. Thanks to my supervisor Steve Burrow, whose patient guidance was a constant source of motivation and clarity, helping me navigate challenges and refine my approach. Martin England, formerly of ICL Rocketry, inspired this project when I saw his incredible E-Reg in operation at EuRoC 2024, which pioneered the concept for UK student teams. His advice and assistance has been invaluable throughout. Thanks to Anthony Healey and Adrian Bot of UoB faculty. Their pressure safety wizardry opened up new avenues for me to pursue in this project, and their insights helped shape a more robust and reliable design. Finally, thanks to everyone on the Hypower team for providing the rest of the system, particularly Oscar Kail who has taken on most of the thankless work iterating valve designs to make this project possible.

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APPENDIX A - E-REGULATOR FLOW EXPERIMENT TEST SEQUENCES AND OPERATIONS

E-Regulator Flow Experiment

Test Sequences and Operations

Jake Marsh

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18/02/2025

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Introduction

The following documents contains details of the operating procedures for the operator of the E-Regulator experiment. See FMEA and Hazard Analysis for risk assessments, and Hypower Teams for software and instrumentation documentation, as well as test checklists to be printed out and used by the operators during the experiment. The Appendix contains verification of the calculations for various attributes of the experiment.

The purpose of these experiments is to develop and test a functioning electronic pressure reducing regulator using an actuated ball valve, for use in a liquid bipropellant rocket.

2 experiments are to be carried out. The first is to characterise the flow performance of the ball valve for various pressure levels and actuation angles. The second is to test the performance of the pressure regulator operating under closed loop control.

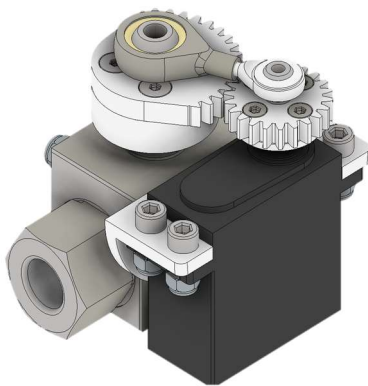


Figure 1 E-Regulator render

High pressure air is supplied by means of a rented BOC Size X compressed air tank at 230 bar, attached via a ¼” hose. These air tanks are typically to supply shielding gas for welding.

A mix of BSPP, NPT and Swagelok fittings are used for attachments. Only one component in the test is experimental – the pressure regulator. This consists of a commercial ball valve with custom servo housing attached allowing it to be remotely operated.

Hence all components in the system are rated to above the maximum possible pressure of 230 bar. If the system is correctly assembled, there is no risk of a catastrophic overpressure.

Nonetheless, it is important that all personnel are familiar with the dangers of and first aid for pressure injection injury. See Appendix 3.1.

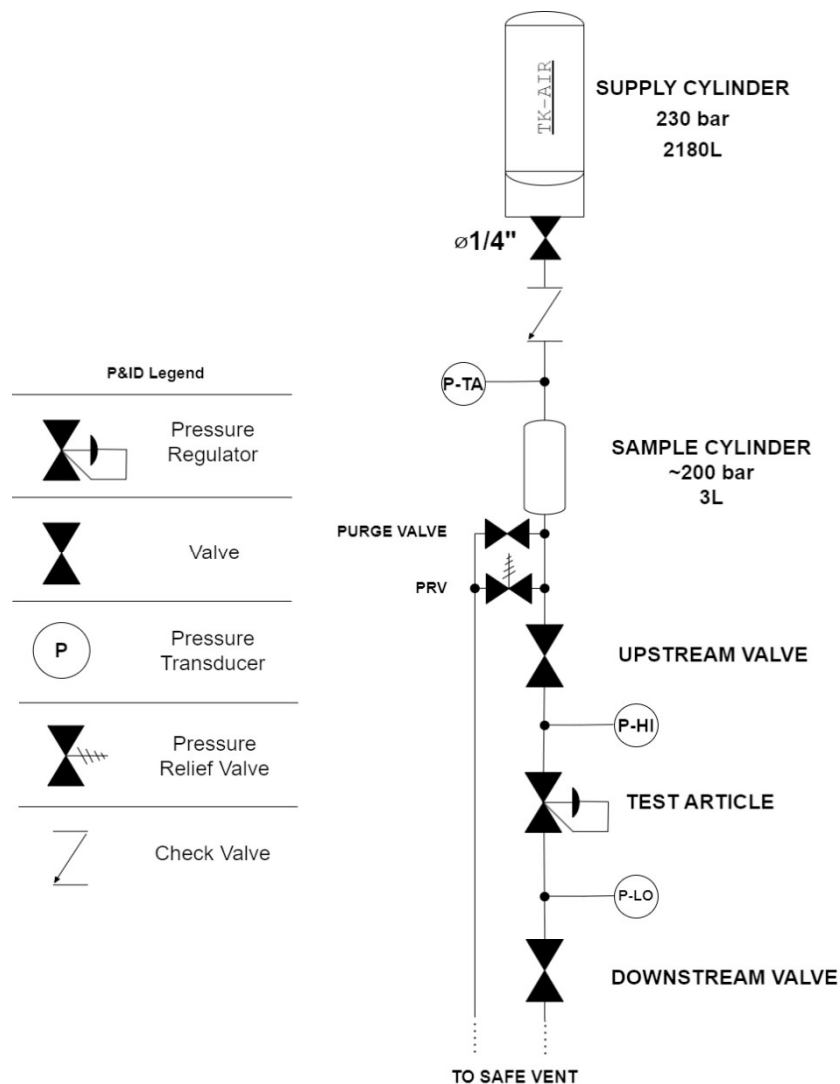
The apparatus should never be pressurised longer than necessary for the test, and 2 operators must be present at all times. Hearing protection is necessary.

Pressure testing at 1.5x MEOP is occurring at an external company, and is required before any testing can be carried out.

	
Dimensions (cm)	94 x 14
Gr. weight (kg)	20
Volume/cylinder contents (m³)	2.4
Cylinder size	X

Figure 2 Air cylinder size

Test Rig



Attribute	Value
Maximum pressure of air in system	230 bar
Lowest pressure rating in system	334 bar
Equivalent air volume at ambient pressure (per test)	0.762 m ³
Total mass of air (per test)	0.784 kg
Equivalent air volume at ambient pressure (total supply)	552 m ³
Maximum Flow Rate ¹	0.63 kg/s
Venting noise ² (at 3 metres)	142 dB
Orifice size (tubing ID)	0.152" / 3.9mm
Flow duration range	1.2 – 7s
Embodied energy (sample tank)	69kJ

¹ using choked flow equation with discharge coefficient 0.9

² According to [University of Southampton](#), at <1 metre distance from vent orifice

1 Test Procedures

1.1 Ambient system inspection and component test actuation

This procedure consists of system inspection by a competent individual, followed by a test actuation of the actively controlled component. This must be performed prior to the commencement of the experiment in Procedure 3.

1. Ensure the system is securely fixed to the frame, and all bolts are fastened.
2. Ensure the frame is securely fixed to the ground, and that the knots are firm.
3. Inspect all pipe fittings are firmly
4. Inspect all non-domed BSPP fittings for dowty washers and firm install.
5. Use the Swagelok Gap Inspection Gauge to ensure all Swagelok fittings are correctly installed.
6. Inspect valve housings for correct meshing of gears and secure housing and cabling.
7. Follow all cables and connectors, and ensure they are plugged in correctly.
8. Inspect all whip checks for strong connection.
9. Ensure that the system is in the **free position**.
10. Connect laptop and power on system.
11. Order close the test article. Check this action completes. **(CLOSE)**
12. Order open the test article. Check this action completes. **(OPEN)**
13. Order the test article to an intermediate angle. Check this action completes. **(TEST)**
14. Ensure area is secure.

1.2 Staged system inspection and component test actuation

This procedure will allow the operators to verify the safety and functionality of the test setup under increasing pressure levels. Test 1.1 is a requirement.

1.2.1 Leak test

1. Ensure the system is in the **pre-leak-test position**.
2. Open the supply needle valve slightly (**DESIRED OPEN**)
3. Once the pressure increases to 2 bara, close the supply valve and start a timer. (**CLOSE**)
4. Inspect closely all connections for noise of leaking. Apply snoop to detect slow leaks.
5. After 15 minutes, check if the pressure has dropped.
6. Rectify any fast or slow leaks, and repeat the test until the system is leak-free.

1.2.2 Staged test actuation

1. Ensure the system is in the **pre-test position**.
2. Open the supply needle valve. (**DESIRED OPEN**)
3. Once the pressure increases to 5 bara, close the supply valve. (**CLOSE**)
4. Check that **P-TA** matches **P-UP**.
5. Ensure that the vent area is clear.
6. Move to a safe distance.
7. Order the test article to open fully. (**OPEN**)
8. Check that the reading on **P-DOWN** is consistent.
9. Manually open the purge vent to depressurise the system. (**OPEN**)
10. Inspect the apparatus for loosening, movement or component failure.
11. Repeat steps 1-9 for pressures 10, 50, 100, 200 bara.

1.3 Pressurisation and flow test procedure

This procedure lists the steps for performing the air flow experiment. Procedures 1.1 and 1.2 are prerequisites.

1. Ensure the area is secure and closed-off.
2. Place the system into the **pre-test position**.
3. Manually open the supply needle valve. **(DESIRED OPEN)**
4. Wait for the pressure reading on **P-TA** to reach 200 bar, then close the supply valve. **(CLOSE)**
5. Ensure the test article is closed.
6. Open the upstream valve. **(DESIRED OPEN)**
7. Move to a safe distance.
8. Ensure the vent area is clear.
9. Begin data logging on all pressure sensors.
10. Order the test article to perform test sequence. The system will now vent. **(TEST)**
11. Allow the system to settle, then end data logging.
12. Place system into the **free position**.

2 State Matrices

This section describes the positions of each valve to be used in each situation.

All valves are manually actuated, with the sole exception of the test article. The operator should ensure they know which direction to turn the handle to open or close each valve.

Most critically, the supply valve should never be opened whilst there is a. open path to vent (via purge, or via the 3 series test valves), as this would allow the supply cylinder to continuously vent to air. The supply valve must be touched only when the system is in the pre-filling position.

Each valve can either be **OPEN**, **CLOSED** or **DESIRED OPEN** (any partially open position the user needs, e.g to slowly pressurise from the cylinder or control the backpressure with the downstream valve).

Free position

Opens the entire system (minus supply cylinder) to atmosphere to prevent stuck pressure.

SUPPLY VALVE	UPSTREAM VALVE	TEST ARTICLE	DOWNSTREAM VALVE	PURGE VALVE
CLOSED	OPEN	OPEN	OPEN	OPEN

Pre-leak-test position

Allows all possible joints to be pressurised to check for leaks.

SUPPLY VALVE	UPSTREAM VALVE	TEST ARTICLE	DOWNSTREAM VALVE	PURGE VALVE
CLOSED	OPEN	OPEN	CLOSED	CLOSED

Pre-test position

Flow is restricted only by the test article, allowing the test to be performed.

SUPPLY VALVE	UPSTREAM VALVE	TEST ARTICLE	DOWNSTREAM VALVE	PURGE VALVE
CLOSED	DESIRED OPEN	CLOSED	DESIRED OPEN	CLOSED

Pre-fill position

Safe position from which to fill sample cylinder.

SUPPLY VALVE	UPSTREAM VALVE	TEST ARTICLE	DOWNSTREAM VALVE	PURGE VALVE
CLOSED	CLOSED	CLOSED	DESIRED OPEN	CLOSED

3 Appendix

3.1 Pressure Safety

The British Fluid Power Association offer a short online course on the dangers and consequences of pressure injection injuries. It is recommended that all operators of the experiment undergo this training at [this link](#)³.

3.2 Attribute calculation and verification

3.2.1 Mass Flow Rate

The flow was assumed to be an ideal compressible gas. Worst case assumptions are used, such as $M=1$ for maximum flow rate.

Using the choked mass flow equation⁴, a mass flow rate of **0.63 kg/s** is found.

$$\dot{m} = \frac{Ap}{\sqrt{T}} \sqrt{\frac{\gamma}{R}} \left(\frac{\gamma + 1}{2} \right)^{-\frac{\gamma+1}{2(\gamma-1)}}$$

3.2.2 Sound Intensity during vent

Using semi-empirical valve calculation from H. Heller and P. Franken, adjusted by researchers from the University of Southampton⁵, we find a sound power level at 3m distance of **142dB**.

$$L_W = 145 + 1.5 * 10 \log (\dot{m}_{exit})$$

3.2.3 Peak Venting Force

Assuming the entirety of the embodied energy is converted into kinetic energy, one finds a final velocity of 420m/s. Further assuming this acceleration occurs in one second results in a peak force of 330m/s.

$$69\text{kJ} = \text{K.E}$$

$$V = 419.5\text{m/s}$$

Assuming flow in one second

$$A = 420\text{m/s/s}$$

$$F = ma: \text{Peak force} = 330\text{N}$$

Apply S.F of 2 for **660N**.

³ <https://bfpa.co.uk/online-training-courses/pressure-injection-awareness-training>

⁴ NASA Glenn Research Centre, 2021: <https://www.grc.nasa.gov/www/k-12/airplane/mflchk.html>

⁵ Heller, Franken, 2022: https://generic.wordpress.soton.ac.uk/isvr-new/wp-content/uploads/sites/422/2022/08/Quaranta-Smith_Valve-noise-at-high-pressure-ratios.pdf

APPENDIX B - EXPERIMENT FAILURE MODES AND EFFECTS ANALYSIS

Item	Potential Failure Mode	Potential Effects of Failure	SEV	Potential Causes of Failure	OCC	Current Design Controls (Prevention)	Current Design Controls (Detection)	Current Design Controls (Mitigation)	DET	RPN
High pressure system - upper tank, tubing and components.										
System must withstand and maintain operating pressure of 230 bar.	High pressure tank or tubing fast rupture.	Rapid release of high pressure air of order 250 bar, risking serious injury and destroying test hardware.	10	Uncontrolled overfilling beyond 250 bar, exceeding pressure limit on system, coupled with operator inability to actuate safety valve	1	Successful proof testing. Monitor pressure level during filling using high pressure sensor. Operator must switch off pump once filled; Operator must ensure functionality of purge valve as specified in Test Operations 1.2.	Loud sound and loss of pressure displayed by transducer, rapid emission of air from rupture.	Operator must stand behind designated physical barrier during pressurisation.	1	10
	High pressure tank or tubing slow leak.	Release of high pressure air, invalidating experiment and risking injury.	6	Incorrectly assembled system; component failure	3	Successful proof testing. Perform leak test as specified in Test Operations 1.2 prior to test operation.	Loss of pressure displayed by transducer, snoop bubbles, audible hissing: See Test Operations 1.2.	Operator must stand behind designated physical barrier during pressurisation.	3	54
	High pressure tank fitting fails.	Rapid release of high pressure air of order 230 bar, risking serious injury and destruction of test hardware. Potential for 3L tank to be propelled away from test setup.	10	Incorrectly assembled system; component failure	2	Successful proof testing. Competant person must check system for correct assembly, perform staged test as specified in Test Operations 1.2. Tank mount blocks 3L tank from exiting test setup.	Loud sound and loss of pressure displayed by transducer, rapid emission of air from rupture.	Operator must stand behind designated physical barrier during pressurisation.	1	20
Pressure transducer must accurately measure pressure.	Failure to update pressure information on monitor.	Inability to gather data or prevent system overpressure.	3	Component failure; cable disconnection; software bug.	6	Competant person must check system for correct assembly, perform staged test as specified in Test Operations 1.2.	No pressure data, or erroneous pressure data reported on monitor.	Operator must activate <i>Purge Sequence</i> .	5	90
System must withstand and maintain operating pressure of 65 bar.	Low pressure tank or tubing fast rupture.	Rapid release of high pressure water of order 65 bar, risking serious injury and destroying test hardware. Spillage of water	9	Test article insufficiently reduces upstream pressure, coupled with failure of pressure relief valve to activate	1	Ensure pressure relief valve is operable as specified in Test Operations 1.2.	Loud sound and loss of pressure displayed by transducer, rapid emission of water from rupture.	Operator must stand behind designated physical barrier during pressurisation.	1	9
	Low pressure tank or tubing slow leak.	Spill of water, invalidating experiment and risking mixing with electric cables.	4	Incorrectly assembled system; component failure	3	Perform leak test as specified in Test Operations 1.2 prior to test operation.	Visible water leak, snoop bubbles, audible hissing: See Test Operations 1.2.	Suspend power and data cables.	5	60
	Failure of fill valve to open/close	Inability to pressurise system.	3	Component failure.	4	Competant person must check system for correct assembly, perform staged test as specified in Test Operations 1.3.	Visible inspection of valve, inability to gain pressure in low pressure system.	If pressurisation has begun, must activate <i>Purge Sequence</i> .	4	48
Flow valve must close after fill is completed.	Flow valve fails to close.	Inability to pressurise system.	3	Component failure, operator error.	4	Perform leak test as specified in Test Operations 1.2 prior to test operation.	Loss of pressure displayed by transducer, snoop bubbles, audible hissing: See Test Operations 1.2.	Mitigated by procedure.	3	36
Flow valve must open to allow flow through system.	Flow valve fails to open.	Inability to flow through system, overpressure in lower pressure tank.	4	Component failure, software bug	3	Perform component operation testing as specified in Test Operations 1.1.	No flow into catch tank visible to operator, increase in pressure in lower pressure system.	PRV automatically activates to vent system. Operator should activate <i>Purge Sequence 3.2</i>	5	60
Pressure transducer must accurately measure pressure.	Failure to update pressure information on monitor.	Inability to gather data or prevent system overpressure.	3	Component failure; cable disconnection; software bug.	6	Competant person must check system for correct assembly, perform staged test as specified in Test Operations 1.2.	No pressure data, or erroneous pressure data reported on monitor.	Operator must activate purge sequence to depressurise system.	5	90
Filling system - hose and supply tank.										
Supply tank must pressurise system to 230 bar.	Seal failure	Release of limited volume of high pressure air >230 bar, potentially causing injury and damaging test hardware.	6	Component failure in filling system	2	Ensure suitable supply cylinder is selected and proper maintenance is carried out by a competent person according to manufacturer specifications.	Emission of air from pump seal, loud sound and loss of pressure.	Operator must stand behind designated physical barrier during pressurisation.	2	24
Filling system must maintain operating pressure of 230 bar.	Pump or tubing fast rupture.	Rapid release of high pressure air up to 230 bar, risking serious injury and destroying test hardware.	10	Exceeding pressure limit on system, coupled with operator inability to actuate safety valve	1	Monitor pressure level during filling using high pressure sensor. Operator must switch off pump once filled; Operator must ensure functionality of purge valve as specified in Test Operations 1.2.	Loud sound and loss of pressure displayed by transducer, rapid emission of air from rupture.	Operator must stand behind designated physical barrier during pressurisation.	1	10
	Pump or tubing slow leak.	Release of high pressure air, invalidating experiment and risking injury.	6	Incorrectly assembled system; component failure	3	Perform leak test as specified in Test Operations 1.2 prior to test operation.	Loss of pressure displayed by transducer, snoop bubbles, audible hissing: See Test Operations 1.2.	Operator must stand behind designated physical barrier during pressurisation. Operator must activate <i>Purge Sequence</i> .	3	54
Hose must maintain coupling at operating pressure of 230 bar.	Loss of coupling.	Release of high pressure air, hose whipping risking injury.	7	Insufficiently tightened fitting; component failure	2	Competant person must check system for correct assembly, perform staged test as specified in Test Operations 1.2; Ensure whip check is fastened.	Hose disconnection and whipping, loss of pressure.	Whip checks fitted to hosing to prevent whipping from going near personnel.	1	14
Test article - actuated ball valve.										
Ball valve must reduce pressure in the lower pressure system to a maximum of 65 bar.	Test article opens too quickly.	High increase in pressure in low pressure system.	1	Ball valve incorrectly set; component failure; controller logic error	10	Robust testing of control system in computational environment, theoretical calculation of valve flow rates prior to deciding test valve angles.	Pressure on display exceeds 65 bar.	System is rated to withstand this	2	20
	Failure to open to any amount.	Inability to flow from high to low pressure system.	3	Component failure; software bug; loose cable	7	Robust testing of software in computational environment, inspection as in Test Operations 1.1.	No pressure drop in high pressure system, no pressure increase in low pressure system.	Operator must activate <i>Purge Sequence 3.2</i> to return system to ambient pressure.	6	126
Ball valve must be able to close to facilitate filling process.	Failure to close.	Lack of isolation of high pressure system during filling.	2	Component failure; software bug; loose cable	7	Robust testing of software in computational environment, inspection as in Test Operations 1.1.	No increase in pressure during filling process	No mitigation required.	6	84
Structure - Frame mount, vent lines and physical barrier.										
Vent line must point towards safe vent area.	Person occupies vent area.	Risk of injury to anyone in vent area if venting is required.	10	Insufficient signage warning public of hazardous operation.	2	Ensure vent area is blocked off and warning is given to owner of space; signage is placed around area discouraging entry.	Operator has line-of-sight on vent area at all times.	Operator must prevent activation of vent valve until area vacated.	5	100
	Vent line no longer directs to the safe vent area.	In event of venting, working fluid could be directed towards the test setup, machinery or personnel.	9	Structural failure of vent line mount.	2	Robust design and inspection of vent line as in Test Operations 1.1.	Operator has line-of-sight on vent area at all times.	Operator can actuate flow valve for alternative vent direction if required.	5	90
Frame structure must support test setup throughout operation.	Test setup collapses, falls over or becomes loose from ground.	If pressurised, could cause catastrophic rupture and depressurisation.	8	Structural failure of frame or tank mount	3	Robust design and inspection of vent support structure as in Test Operations 1.1.	Operator has line-of-sight on test setup at all times.	Operator must activate <i>Purge Sequence</i> to return system to ambient pressure as quickly as possible.	1	24
Electronics - Cabling for power and signals										
Cables must provide power and signals to all components throughout operation.	Cables fail to deliver power or signals.	Instrumentation failure, inability to actuate test article.	5	Cable pulled out; component failure.	4	Competant person must check system for correct assembly, perform component test as specified in Test Operations 1.1 and 1.2.	Loss of power or instrumentation will cause alarm to sound.	Operator must activate <i>Purge Sequence</i> to return system to ambient pressure.	2	40
Sound - Risk from venting noise										
Noise protection must protect personnel from bursts of sound up to 142 dB.	Hazard occurs from normal operation.	Hearing damage	8	Failure to wear proper hearing protection.	8	N/a	N/a	Wear hearing protection rated in excess of 40 SNR.	1	64

SEV - Severity	
10	Severe risk of injury or death without warning.
9	Severe risk of injury to operator with warning.
8	Risk of injury to operator, loss of test setup.
7	Damage to multiple components in test setup, risk of injury to operator
6	Failure of experiment, damage to components in test setup, risk of injury to operator
5	Failure of experiment, risk of damage to components in test setup
4	Failure of experiment, requiring purge and refilling.
3	Impact on validity of data gathered, or delay in process.
2	Potential impact on validity of data gathered, or short delay in process,
1	No discernable effect.

OCC - Occurance	
10	> 1 in 10
9	> 1 in 20
8	> 1 in 50
7	>1 in 100
6	>1 in 500
5	>1 in 1000
4	>1 in 2000
3	>1 in 10,000
2	>1 in 100,000
1	Failure is eliminated by prevention

DET - Detection	
10	Impossible to detect
9	Failure mode and cause is not easily detected
8	Failure is detectable by monitoring of instrumentation by operator.
7	Failure is detectable by operator by subtle visual or audible means.
6	Failure is obvious by monitoring of instrumentation by operator.
5	Failure is detectable by operator by visual or audible means.
4	Failure is detectable by inspection of test setup, or as part of Test Operation 1.2 (proof test)
3	Failure is detectable by inspection of test setup, or as part of Test Operation 1.1 (leak test)
2	Alarm will automatically signal failure to operator.
1	Failure is obvious, e.g very loud.

APPENDIX C - EXPERIMENT HAZARD AND OPERATIONS ANALYSIS

Hazard	Cause	Effect	Probability	Corrective or Preventative Measures
Pressure release up to 230bar towards personnel	Improper assembly or operation resulting in high pressure air directed at operator.	Injection injury, severity depending on proximity and pressure.	Remote	All components rated higher than highest possible pressure. Assembly is checked by competent technician. Ensure vent area is clear and personnel are at safe distance.
Loud venting noise up to 142dB	Shock and mixing of vented flow causing a pressure wave.	Hearing damage to operators or anyone nearby without hearing protection. Potential for secondary effects.	Frequent	All personnel to wear ear protection rated in excess of 40 SNR, vent location to be placed in attenuating location. Surroundings must be cleared.
Impulse from vent launches hardware.	Improperly secured venting apparatus.	Kinetic injury from fast-moving loose components. Pressure injection.	Remote	Force from venting has been quantified and venting apparatus is rated to in excess of this by a safety factor of 2.
Fitting projection	Poorly tightened fitting fails under pressure.	Kinetic injury from fast-moving loose components. Pressure injection.	Remote	Assembly checked by competent technician, all fittings rated in excess of highest possible pressure.
Loose hose venting and whipping	Improper assembly or operator error when disconnecting hose.	Kinetic injury from whipping hose. Also risk of injection injury.	Improbable	Rated whip checks connected to all hose connections.
Loose tank flight.	Failed tank connection under pressure causes tank to disconnect and fly around.	Injection injury, kinetic injury from fast-moving tank.	Remote	Assembly checked by competent technician, test apparatus secures tank to ground.
Brittle failure	Explosion of sample pressure vessel	Kinetic injury from debris, pressure injection, blast.	Remote	All components rated higher than highest possible pressure.
Secondary failures	Failure triggered by primary blast.	Kinetic injury from debris e.g glass shards.	Improbable	Position all personnel safely away from clear secondary failure points, ensure the room is suitable.

Occurance per test	
Frequent	Likely to occur during test
Probable	>1 in 10
Occasional	> 1 in 50
Improbable	>1 in 500
Remote	<1 in 500